Design Process Planning and Management

Lab manual for

EX-04: Thermal analysis of simplified engine cylinder

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EX04: Thermal analysis of simplified engine cylinder with cooling fins

Target Date: 15 Feb

AIM

Perform steady state thermal analysis of engine cylinder with two different fin configuration and Compare the maximum and minimum temperature and temperature distribution for the given thermal environment

Tool : Fusion 360

Approach: Finite element simulation

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Simulation Input

Simplified model of engine cooling fin around cylinder. Two fin configurations: a) Rectangular b) Trapezoidal

Geometric dimensions are in mm

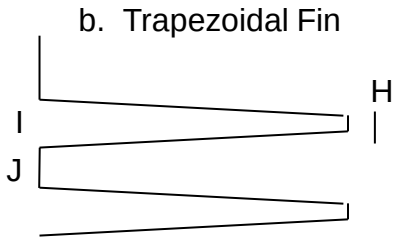
Inside cylinder surface is exposed to combustion with 1200 c

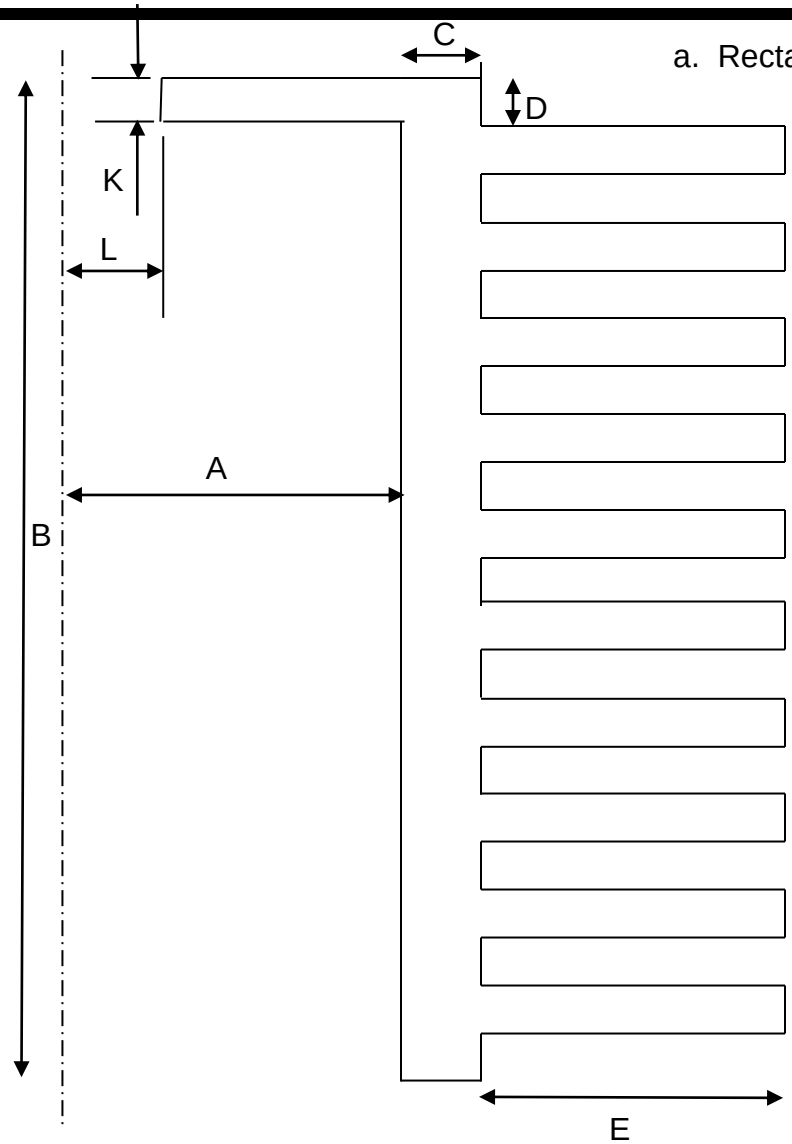
Convective heat transfer coeff at inner cylindrical surface is 400 w/m2. c

All external cylinder and fin surface has conv heat transfer coefficient of 50 W/m2/c and 30 c

Emissivity of all surface : 0.1

Material Aluminium 6061





A= 50 mm
B=75 mm
C= 5 mm
D= 7.5 mm
E= 45 mm
F=3 mm
G=3 mm
H=1.5 mm
I=3 mm
J=2 mm
K= 5 mm
L= 7.5 mm
X=10 Nos
Y=12 Nos

Number of fins = X for rectangular fin =Y for trapezoidal fin

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Procedure

1. Make a sketch to the dimensions shown in figure**. Note create geometric model in 'm'**
2. Revolve the sketch and make body of cylinder with fin (X Nos)
3. Save model EX04_ Roll No
4. Go to simulation menu
5. Make analysis setup – Steady state thermal analysis
6. Assign Aluminium material to the geometric body
7. Define loads
   - Assign convective load to the inner surface
   - Assign convective load to all other surface
   - Assign radiation load with emissivity to internal surfaces and all external surface with respective temperature

8 Define meshing parameter with the average size as **4 mm**
9 Generate mesh
10 Check the model for input definition completeness
11 Solve the model
12 Post process the model to show the thermal gradient, max and minimum temperature
13 Save model
14 Rerun the analysis with element size **2 mm** and observe for change in max and min temperature
15 Save model
16 Repeat the steps 1-12 with Y number of fins of trapezoidal shapes
17 Save model
18 Compare the temperature distribution for rectangular configuration with two different mesh sizes
19 Compare temperature distribution between rectangular and trapezoidal fin configuration'
20 Tabulate the maximum and minimum temperature in the cylinder and location

Upload the PDF of this completed report in Fusion 360 and VTOP.
All the models version to be stored in your folder

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Table 1: Thermal simulation results summary

Analysis case	Fin Type	Global element size	Max		Min	
			Temp (c)	Location	Temp (c)	Location
1	Rectangular	4	701.9	Near the top inner ring.	476.5	At the edge of the fins.
2	Rectangular	2	702.5	At the bottom of the top ring.	476.3	At the edge of the fins.
3	Trapezoidal	2	663.7	At the bottom of the top ring.	282.8	At the edge of the fins.

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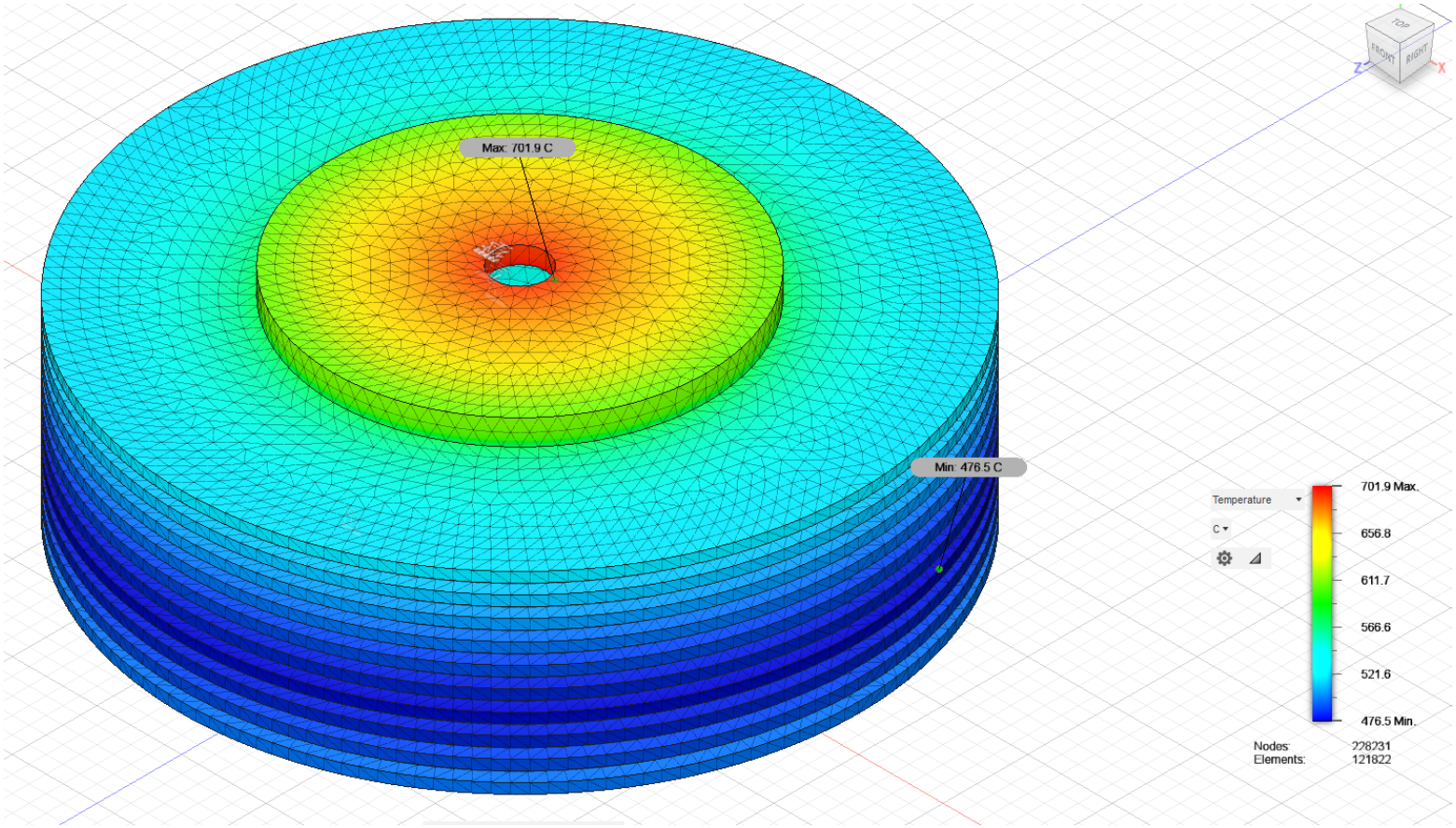
Figures: Temperature distribution for all analysis case



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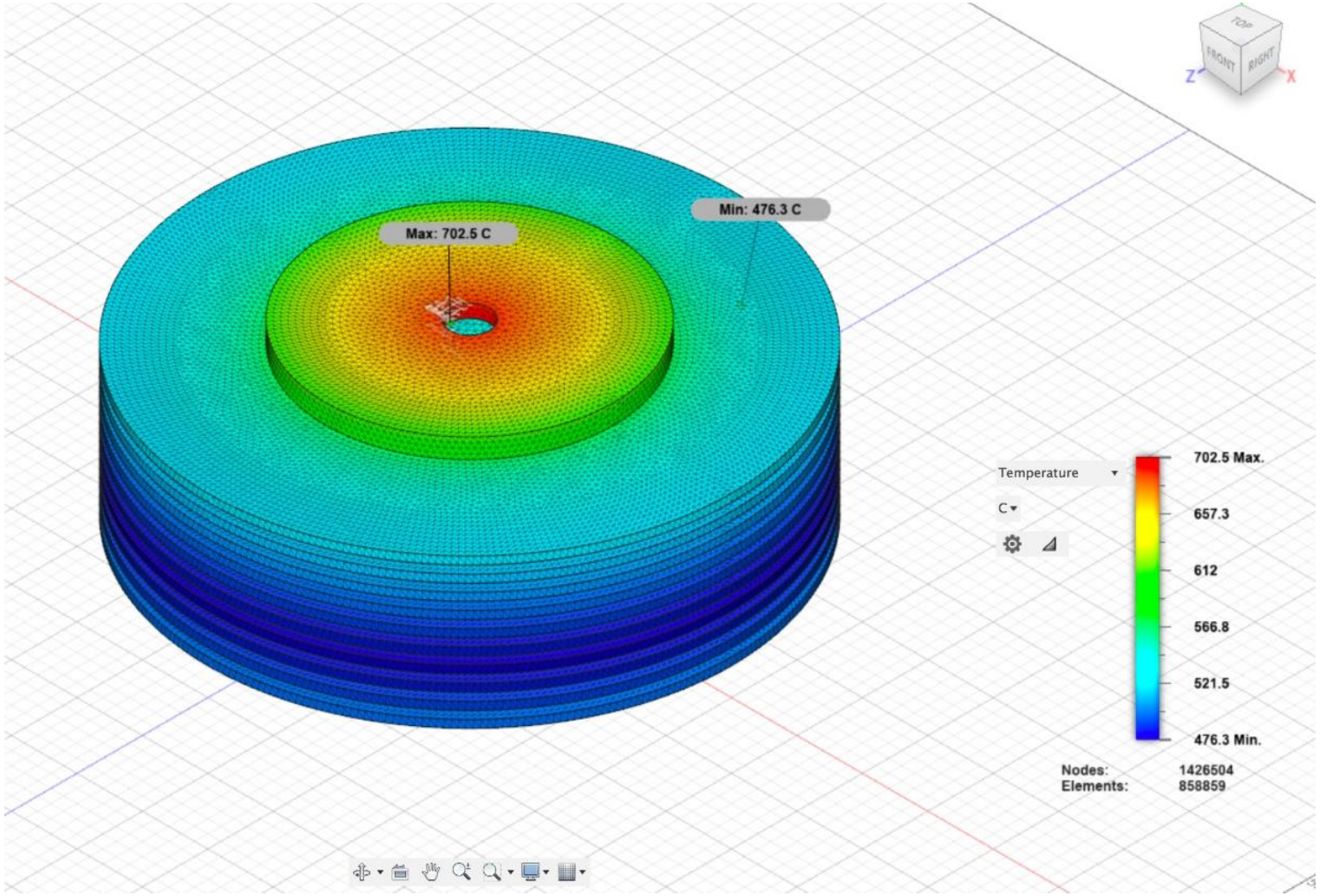

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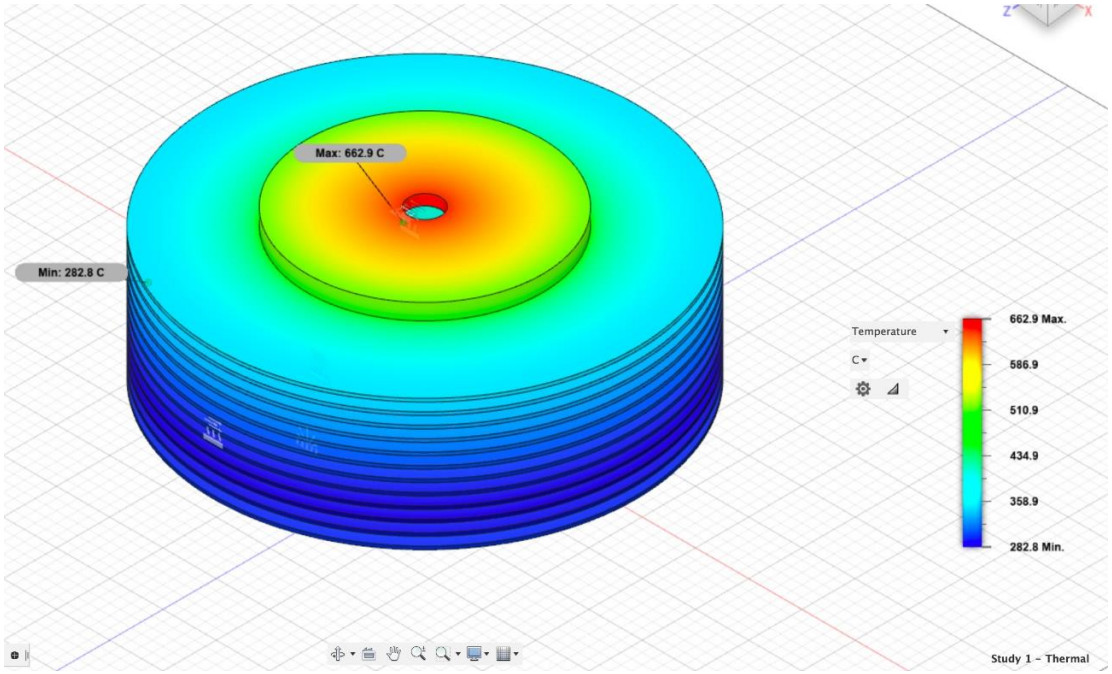

Conclusion

- The values were consistent through out all the readings.
- The max value was found to be 702.5 C and the min temp. was 476.3 C at the edge of the 7th and 8th for the rectangular fins.
- The max value for trapezoidal fins was 663.7 C and the min value was 282.8 C.
- The trapezoidal fins performed better than the rectangular fins.